β turns

These are secondary structural elements with four amino acids that can reverse (turn) the direction of a polypeptide and thus help the polypeptide to form a globular shape. They are mostly found on the surface of proteins. The amino acids proline and glycine are more frequently found in β turns. They are also mostly found to connect two different α helices or β strands to form super secondary structure motifs such as helix-turn helix, beta meander, beta barrel etc.

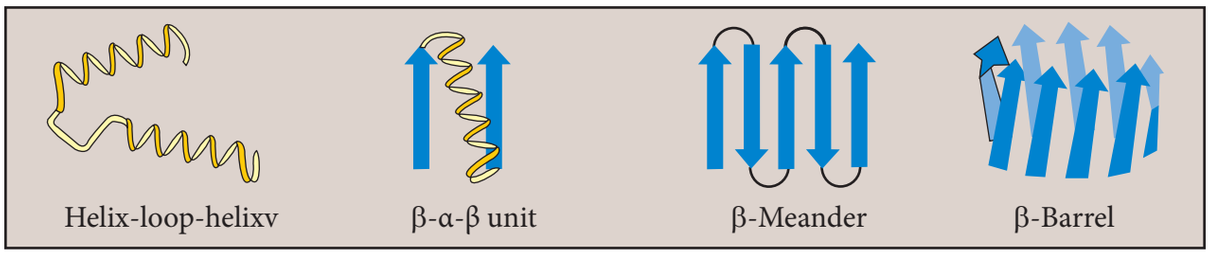


Figure 3.11 The secondary structure elements connected through β turns

3.3.3 Tertiary structure

The polypeptide folds in such a way that the secondary structure elements are packed compactly to form an overall three-dimensional structure called its **tertiary structure**. The tertiary structure is stabilized mainly by the interactions between the R groups (side chains) of the amino acids.

The interactions that contribute to tertiary structure are hydrogen bonds, ionic interactions, dipole-dipole interactions and Vander Waals Forces. The above mentioned interactions are also known as non bonded interactions.

The side chains with like charges such as Lys and Arg repel one another, while those with opposite charges such as Lys and Asp can form an ionic interaction. Similarly, polar R groups can form hydrogen bonds and other dipole-dipole interactions.

The amino acids with non polar, hydrophobic R groups cluster together on the inside of the protein through hydrophobic interactions. This cluster is also known as the hydrophobic core and it is an important feature of globular proteins. Similarly, the hydrophilic amino acids, (i.e.) the amino acids with side chains containing charged groups are present on the surface of globular proteins to interact with surrounding water molecules.

The sulphur containing side chains of two cysteine residues can form a covalent bond known as a **disulfide bond**. The disulfide bonds help to bring together two different

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